

Dupire v.s. Neural Networks Delta Hedging

Erdos Project

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- Dupire model: Why constant volatility is insufficient and how local volatility uses the implied volatility surface.
- Neural network approach: How a learning-based method approximates option hedging behavior from market features.
- Results and comparison: How the Dupire and neural-network hedges perform in the backtest.

Motivation: Beyond Black–Scholes

- Black–Scholes dynamics:

$$dS_t = rS_t dt + \sigma S_t dW_t.$$

- However, in reality volatilities vary across strikes and maturities:

$$\sigma_{\text{imp}} = \sigma_{\text{imp}}(K, T).$$

- This implied volatility surface shows that one constant σ cannot fit all traded option prices.
- Dupire starts from this market surface and asks how to build a dynamic model consistent with it.

Dupire Local Volatility

- Replace constant volatility with a state- and time-dependent function:

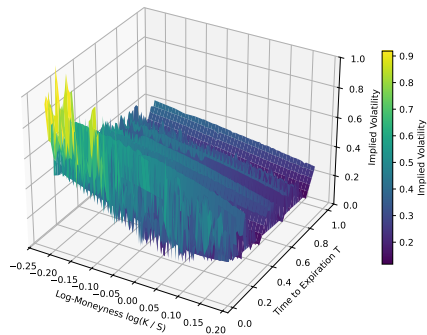
$$dS_t = rS_t dt + \sigma_{\text{loc}}(S_t, t)S_t dW_t.$$

- Financially, $\sigma_{\text{loc}}(S, t)$ is the instantaneous volatility assigned to state (S, t) .
- Let $C(K, T)$ be the time-0 price of a European call with strike K and maturity T .
- Dupire's formula recovers local variance from derivatives of the call price surface:

$$\sigma_{\text{loc}}^2(K, T) = \frac{\frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K}}{\frac{1}{2}K^2 \frac{\partial^2 C}{\partial K^2}}.$$

- Here C is the call price, K is strike, T is maturity, and r is the risk-free rate.

From Implied to Local Volatility

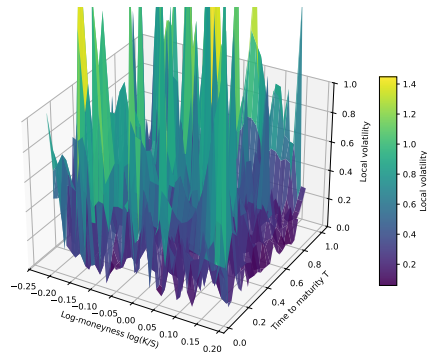


Implied volatility surface

$$\sigma_{\text{imp}}(K, T)$$



$$\sigma_{\text{loc}}^2(K, T) = \frac{\frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K}}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}}$$



Local volatility surface

$$\sigma_{\text{loc}}(S, t)$$

Delta Proxy for Hedging

- A fully model-consistent Dupire delta would price the option under local-volatility dynamics and then compute

$$\Delta_{\text{Dupire}} = \frac{\partial V}{\partial S}.$$

- In our implementation, we use a practical proxy: look up $\sigma_{\text{loc}}(S, t)$ and plug it into the Black–Scholes delta formula.

$$\Delta_{\text{proxy}} = N\left(\frac{\log(S/K) + (r + \frac{1}{2}\sigma_{\text{loc}}^2) T}{\sigma_{\text{loc}} \sqrt{T}}\right).$$

- This gives a fast, interpretable hedge ratio for daily rebalancing.
- As a check, a finite-difference Monte Carlo Dupire delta gave a similar order of magnitude: 0.47 versus proxy value 0.54 for the ATM example.

Interpretation and Limitations

- Under ideal smooth, arbitrage-free data, Dupire calibrates exactly to the observed call price surface.
- It turns market-implied option prices into a dynamic stock-price model for pricing and hedging.
- In practice, data are finite and noisy: missing quotes, bid-ask spreads, and interpolation matter.
- The formula uses numerical derivatives, so noise can be amplified.
- This motivates smoother approximations and learning-based methods, which the next section considers.